

STEEL: Singularity-aware Reinforcement Learning

Abstract

We propose a new batch RL algorithm to learn an optimal policy in a nonparametric setting of infinite-horizon Markov decision process with continuous states and actions, where the distributions induced by unknown optimal policies can have non-overlapping regions with that of the offline data generating distribution (i.e., singularity in both state and action spaces). We call our algorithm STEEL: Singularity-aware REinforcement Learning. The algorithm is motivated by our novel finite-sample error analysis of off-policy evaluation (OPE) via Lebesgue decomposition. We use maximum mean discrepancy and distributionally robust optimization to characterize the OPE error caused by possible singularity and to enable nonparametric smooth extrapolation. Under pessimism and nonparametric smooth function class conditions, we derive the first finite-sample regret guarantee for our algorithm allowing for singularity. Compared with existing algorithms that require overlapping regions between the distribution induced by an unknown optimal policy and the offline data generating distribution, our STEEL greatly improves the applicability and robustness of batch RL. In addition, we propose a two-step adaptive STEEL, which is nearly tuning-free, to reduce the regret error further. Extensive simulation studies and one (semi)-real experiment on personalized pricing demonstrate the superior performance of our methods in dealing with general distributional mismatch in batch RL.

1 Introduction

We study batch reinforcement learning (RL), which aims to learn an optimal policy using the pre-collected data without further interacting with the environment, and without knowing the functional forms of per period reward function and the Markov transition probabilities. Arguably, the biggest challenge for solving such a problem nonparametrically is the distributional mismatch between the batch data (generating) distribution and those induced by the unknown optimal policies (Levine et al., 2020).

To address the issue of distributional mismatch in offline RL, prior works require *absolute continuity* assumption, e.g., there does not exist a non-overlapping region between the batch data distribution and the distribution induced by target policies. For example, classical

methods such as [Antos et al. \(2008\)](#) require the batch data to contain every possible state-action pair. Recent literature (e.g., [Kumar et al., 2019](#); [Jin et al., 2021](#)) has found that, by incorporating the idea of pessimism, it suffices to require the batch data contain state-action pairs induced by an optimal policy. However, this weaker assumption of absolute continuity of the distribution induced by an unknown optimal policy with respect to the data distribution is still difficult to check a priori and can be easily violated. In such a case, *singularity* arises; see [Section 2](#) for the formal definitions of absolute continuity and singularity between any two probability measures. Indeed, in real world applications with multi-dimensional continuous state and continuous action spaces, there often exist non-overlapping regions between the batch data distribution and the target distributions induced by the optimal policies, and all aforementioned methods fail.

To address the issue of singularity, in this paper, we propose a new sample efficient and singularity-aware policy-iteration-type algorithm for finding an optimal policy under the framework of an infinite-horizon Markov decision process (MDP) with continuous states and actions. By saying a sample efficient algorithm, we refer to that the sample complexity requirement in finding an optimal policy is polynomial in terms of key parameters ([Zhan et al., 2022](#)). We call our algorithm STEEL: Singularity-aware Reinforcement Learning.

Our algorithm is motivated by a simple but novel finite-sample error analysis on off-policy evaluation (OPE). The goal of OPE is to use the batch data for evaluating/estimating the performance of any candidate policies ([Shi, Zhang, Lu and Song, 2020](#)). By leveraging Lebesgue’s Decomposition Theorem, we decompose the error of OPE into two parts: the absolutely continuous part and the singular part with respect to the data distribution (induced by the behavior policy). The OPE error of the absolutely continuous part can be calibrated by the data distribution using the change of measure, and then be controlled by applying existing OPE methods (such as the ones allow for pessimism). To control the

OPE error of the singular part, we use distributionally robust optimization to characterize its worst-case error from the data distribution measured by an integrated probability metric (IPM), such as maximum mean discrepancy (MMD), total variation distance and Wasserstein metrics. Once we understand these two sources of the OPE error induced by any given target policy, a new estimating method for OPE without requiring the absolute continuity can be formulated, based on which a policy iteration algorithm with pessimism is proposed to learn/estimate an optimal policy. In addition, we propose a two-step adaptive STEEL algorithm that further reduces the regret error bound and is almost tuning-free.

From the theoretical perspective, we show that under some technical conditions, the regret of our estimated policy converges to zero at a satisfactory rate in terms of total decision points in the batch data even when singularity arises. This novel result demonstrates that our method can be more applicable in solving batch RL problems, in comparison to existing solutions. More specifically, when the distribution induced by the optimal policy is covered by our batch data generating process in the usual manner, we recover the existing theoretical results given by those pessimistic RL algorithms (e.g., [Xie et al., 2021](#); [Fu et al., 2022](#)). In contrast, when absolute continuity fails, our algorithm is provable to find an optimal policy with a finite-sample regret warranty. The success of our proposed method relies on a stronger nonparametric smooth function class assumption, i.e., Bellman completeness with respect to a smooth function class, which provides a desirable extrapolation property so that the singular part incurred by the distributional mismatch can be properly controlled.

Our work can also provide theoretical guidance for other existing algorithms, such as [Silver et al. \(2014\)](#), aiming at a deterministic optimal policy in a multi-dimensional continuous action space with covariates' shifts in a multi-dimensional state space. In contrast, existing methods can only allow for singularity in certain state or action regions and still

require absolute continuity to some extent. For example, [Chen et al. \(2016\)](#); [Kallus and Uehara \(2020\)](#) leverage kernel smoothing techniques to address the issue of singularity on the action space only, which is known to suffer from the curse of dimensionality with multi-dimensional continuous action spaces.

Our algorithms and finite-sample regret guarantee are new even when they are specialized to the contextual bandit problems; see [Section 3.1](#). To demonstrate the effectiveness of our new algorithms, we conduct extensive simulation studies using a contextual bandit problem with multi-dimensional state and action spaces. We show that compared with two existing baseline methods, under a possible singularity, our algorithm has a significantly better finite-sample performance. In particular, we observe that our algorithm converges faster and is more robust when the singularity arises, and that our adaptive STEEL algorithm has superior performance over the non-adaptive version. Finally, we also apply our method to a personalized pricing application using the data from a US auto loan company, and find that our method outperforms the existing baseline methods as well.

2 MDP Framework and Notation

In this section, we briefly introduce the framework of the discrete-time homogeneous MDP, which is the foundation of many RL algorithms, and the related notations ([Sutton et al., 1998](#)). Consider a time series $\{S_t, A_t, R_t\}_{t \geq 0}$, where at the decision point t , S_t is the state of the environment, A_t is the action, and $R_t := \tilde{R}(S_t, A_t, S_{t+1})$ is the immediate reward the agent receives. We let $\mathcal{S}$ and $\mathcal{A}$ respectively denote the state and action spaces. In this paper, we assume both $\mathcal{S} \subseteq \mathbb{R}^{d_S}$ and $\mathcal{A} \subseteq \mathbb{R}^{d_A}$ are continuous and multi-dimensional. Let $\mathcal{B}(\mathcal{S})$ be a family of Borel subsets of $\mathcal{S}$, and $\mathcal{A} = \bigotimes_{i=1}^{d_A} \mathcal{A}_i$ with $d_A \geq 1$, where $\mathcal{A}_i$ is the space for i -th coordinate of the action. Under the setup of MDP, the following two assumptions are standard.

Assumption 1 *There exists an unknown time-invariant transition kernel P such that for every $t \geq 0$, $s \in \mathcal{S}$, $a \in \mathcal{A}$ and any measurable set $F \in \mathcal{B}(\mathcal{S})$,*

$$\Pr(S_{t+1} \in F \mid S_t = s, A_t = a, \{S_j, A_j, R_j\}_{0 \leq j < t}) = P(S_{t+1} \in F \mid S_t = s, A_t = a),$$

where $\{S_j, A_j, R_j\}_{0 \leq j < t} = \emptyset$ if $t = 0$. In addition, for each $(s, a) \in \mathcal{S} \times \mathcal{A}$, $P(\bullet \mid s, a)$ is absolutely continuous with respect to the Lebesgue measure, with a probability density function $q(\bullet \mid s, a)$.

We note that one can test this time-homogeneous first-order Markov Assumption 1 using the batch data; (e.g., [Shi, Wan, Song, Lu and Leng, 2020](#); [Li et al., 2022](#)).

Assumption 2 *For any $t \geq 0$, the immediate reward $R_t = \tilde{R}(S_t, A_t, S_{t+1})$ is a measurable function of (S_t, A_t, S_{t+1}) , where the functional form $\tilde{R} : \mathbb{R}^{2d_S + d_A} \rightarrow \mathbb{R}$ is unknown. In addition, there exists a finite constant $R_{\max}$ such that $\sup_{t \geq 0} |R_t| \leq R_{\max}$.*

The uniformly bounded immediate reward Assumption 2 is imposed to simplify the technical proofs and can be replaced by finite higher order moment assumptions on R_t . Under Assumptions 1 and 2, the reward function at time point t $r(S_t, A_t) := \mathbb{E}[R_t \mid S_t, A_t]$ is well-defined, and its functional form is unknown but does not depend on $t \geq 0$.

An essential goal of RL is to find an optimal policy that maximizes some utility. In this paper, we use expected discounted sum of rewards to measure the performance of each policy. A policy is defined as a decision rule according to which an agent chooses her action A_t based on the environment state S_t at each decision point t . We focus on the stationary policy π , which is a vector-valued function mapping from the state space $\mathcal{S}$ into a probability distribution over the action space $\mathcal{A}$. Specifically, $\pi = \bigotimes_{i=1}^{d_A} \pi_i$, where π_i is a function mapping from $\mathcal{S}$ into a distribution over $\mathcal{A}_i$. The sufficiency of focusing on the stationary policies is guaranteed by Assumptions 1 and 2; see Section 6.2 of [Puterman \(1994\)](#) for the justification. We note that stationary policy does not rule out the deterministic policy,

which is a function mapping from the state space $\mathcal{S}$ into the action space $\mathcal{A}$. We evaluate the overall performance of a policy π by the so-called *policy value* defined as

$$\mathcal{V}(\pi) := (1 - \gamma) \mathbb{E}_{S_0 \sim \nu} \left[\sum_{t=0}^{\infty} \gamma^t R_t \right], \quad (1)$$

where $\gamma \in [0, 1)$ is a known time discount rate, and ν is some *known* reference distribution for the initial state S_0 . Here $\mathbb{E}_{S_0 \sim \nu}[\cdot]$ refers to the expectation that all actions along the trajectory follow the stationary policy π , and then average over S_0 using ν . Our goal in this paper is to search for an *in-class* optimal policy π^* such that

$$\pi^* \in \arg \max_{\pi \in \Pi} \mathcal{V}(\pi), \quad (2)$$

where $\Pi = \bigotimes_{i=1}^{d_A} \Pi_i$ and Π_i is some pre-specified class of stationary policies for i -th coordinate of $\mathcal{A}$. Some commonly used policy classes include linear decision functions, neural networks and decision trees.

Under Assumptions 1 and 2, the policy value (i.e. utility) function $\mathcal{V}(\cdot)$ is well-defined but its functional form is unknown (due to the unknown functional form of $P(\bullet | s, a)$). We now introduce additional definitions and notation to facilitate the estimation/learning of $\mathcal{V}(\bullet)$ and π^* from batch data. For each stationary policy π , we define a Q -function to measure its performance starting from any state-action pair $(s, a) \in \mathcal{S} \times \mathcal{A}$, denoted by

$$Q^\pi(s, a) := \mathbb{E}^\pi \left[\sum_{t=0}^{\infty} \gamma^t R_t \mid S_0 = s, A_0 = a \right]. \quad (3)$$

For any generic function Q defined over $\mathcal{S} \times \mathcal{A}$, we let $Q(s, \pi(s)) := \int_{a \in \mathcal{A}} Q(s, a) \pi(a | s) da$. Then the utility function $\mathcal{V}(\pi)$ can be equivalently expressed as

$$\mathcal{V}(\pi) = (1 - \gamma) \mathbb{E}_{S_0 \sim \nu} [Q^\pi(S_0, \pi(S_0))]. \quad (4)$$

We define a discounted visitation probability measure over $\mathcal{S} \times \mathcal{A}$ induced by a policy π as

$$d_\gamma^\pi := (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t q_t^\pi, \quad (5)$$

where q_t^π is the marginal probability measure of (S_t, A_t) induced by the stationary policy π with the initial state distribution ν . Then $\mathcal{V}(\pi)$ can be equivalently expressed as

$$\mathcal{V}(\pi) = \int_{\mathcal{S} \times \mathcal{A}} r(s, a) d_\gamma^\pi(ds, da) = \mathbb{E}_{S_0 \sim \nu} [r(S_0, \pi(S_0))], \quad (6)$$

where $r(s, \pi(s)) := \int_{a \in \mathcal{A}} r(s, a) \pi(a | s) da$.

We focus on the batch setting, where the observed data consist of N independent and identically distributed (i.i.d.) copies of $\{S_t, A_t, R_t\}_{t \geq 0}$ up to T decision points. For the i -th trajectory, where $1 \leq i \leq N$, the data can be represented by $\{S_{i,t}, A_{i,t}, R_{i,t}, S_{i,t+1}\}_{0 \leq t < T}$. The following is a standard assumption imposed on our batch data generating process.

Assumption 3 *The batch dataset $\mathcal{D}_N := \{(S_{i,t}, A_{i,t}, R_{i,t}, S_{i,t+1}) : 0 \leq t < T\}_{i=1}^N$ is i.i.d. copies of $\{(S_t, A_t, R_t, S_{t+1}) : 0 \leq t < T\}$ that is generated according to an unknown stationary behavior policy π^b .*

Let $q_t^{\pi^b}$ be the marginal probability measure over $\mathcal{S} \times \mathcal{A}$ at the decision point t induced by the behavior policy π^b , and $d_\gamma^{\pi^b}$ be its corresponding discounted visitation probability. Sometimes we simply call $d_\gamma^{\pi^b}$ the Data Generating Probability (DGP), and use $\bar{\mathbb{E}}[\cdot]$ to denote the expectation with respect to $d_\gamma^{\pi^b}$.

We define an average visitation probability measure across T decision points as

$$\bar{d}_T^{\pi^b} := \frac{1}{T} \sum_{t=0}^{T-1} q_t^{\pi^b}. \quad (7)$$

The corresponding expectation with respect to $\bar{d}_T^{\pi^b}$ is denoted by $\bar{\mathbb{E}}$ and sometimes called the data expectation. Throughout the theoretical analysis below, $\bar{d}_T^{\pi^b}$ is the reference measure used in the empirical Bellman residual bounds and in the kernel operator defined in the supplement.

In this paper, we aim to leverage the batch data to find an in-class optimal policy π^* defined in (2) without assuming strong data coverage condition. In particular, we allow

that the discounted visitation probability measure $d_\gamma^{\pi^*}$ over $\mathcal{Z} := \mathcal{S} \times \mathcal{A}$ is not absolutely continuous with respect to the DGP $d_\gamma^{\pi^b}$. For any two probability measures $\mathbb{Q}, \mathbb{P}$ on $\mathcal{Z}$, we use $\mathbb{Q} \ll \mathbb{P}$ when $\mathbb{Q}$ is absolutely continuous with respect to $\mathbb{P}$ (i.e., for any measurable set F if $\mathbb{P}(F) = 0$ then $\mathbb{Q}(F) = 0$), while $\mathbb{Q} \perp \mathbb{P}$ when $\mathbb{Q}$ and $\mathbb{P}$ are singular (i.e., there exists a measurable set F such that $\mathbb{P}(F) = 0$ but $\mathbb{Q}(F) = 1$). By the Lebesgue Decomposition Theorem (see, e.g., Theorem 6.3 of [Gray \(2009\)](#)), we have for any discounted visitation probability measure d_γ^π on $\mathcal{Z} = \mathcal{S} \times \mathcal{A}$, there are unique probability measures λ_1^π and λ_2^π , and constants $\tilde{\lambda}_1^\pi(\mathcal{Z}) \in [0, 1]$, $\tilde{\lambda}_2^\pi(\mathcal{Z}) = 1 - \tilde{\lambda}_1^\pi(\mathcal{Z})$ such that

$$d_\gamma^\pi = \tilde{\lambda}_1^\pi(\mathcal{Z}) \times \lambda_1^\pi + \tilde{\lambda}_2^\pi(\mathcal{Z}) \times \lambda_2^\pi, \quad \text{with } \lambda_1^\pi \ll d_\gamma^{\pi^b} \text{ and } \lambda_2^\pi \perp d_\gamma^{\pi^b}. \quad (8)$$

In the standard Lebesgue Decomposition Theorem, there are no constants $\tilde{\lambda}_i^\pi$. We have these normalized constants because we restrict λ_i^π to be the probability measures. We will use this decomposition to provide a new OPE for $\mathcal{V}(\pi)$ allowing for singularity, based on which we learn the optimal policy π^* . The use of such a decomposition can split the state-action space into the overlapping and non-overlapping regions between the data distribution and the distribution induced by the target policy. In this case, the distributional shift on the overlapping region can be handle by the existing methods, while the non-overlapping part requires a special treatment, where we use the idea of the distributionally robust optimization to quantify the worst case error of the OPE, which is our main contribution.

To allow for comparison between any two distributions that are not absolutely continuous with respect to each other, we use an integrated probability metric (IPM) between two probability distributions $\mathbb{P}$ and $\mathbb{Q}$ on $\mathcal{Z}$, defined as ([Müller, 1997](#))

$$\tau_\Xi(\mathbb{P}, \mathbb{Q}) := \sup_{f \in \Xi: \|f\|_\Xi \leq 1} |\mathbb{E}_{Z \sim \mathbb{P}}[f(Z)] - \mathbb{E}_{Z \sim \mathbb{Q}}[f(Z)]|, \quad (9)$$

where Ξ is a class of real-valued measurable functions. By definition we have that for any

$f \in \Xi$,

$$|\mathbb{E}_{Z \sim \mathbb{P}}[f(Z)] - \mathbb{E}_{Z \sim \mathbb{Q}}[f(Z)]| \leq \tau_{\Xi}(\mathbb{P}, \mathbb{Q}) \times \|f\|_{\Xi}.$$

By different choices of Ξ , the IPM family of probability metrics includes many widely used examples, such as maximum mean discrepancy (MMD), total variation distance (TV) and Wasserstein metric (W_p) with an integer $p = 1$. In particular, the MMD between $\mathbb{P}$ and $\mathbb{Q}$ is defined as

$$\text{MMD}_k(\mathbb{P}, \mathbb{Q}) := \sup_{f \in \mathcal{H}_k: \|f\|_{\mathcal{H}_k} \leq 1} |\mathbb{E}_{Z \sim \mathbb{P}}[f(Z)] - \mathbb{E}_{Z \sim \mathbb{Q}}[f(Z)]|, \quad (10)$$

where $\mathcal{H}_k$ is a reproducing kernel Hilbert space (RKHS) with the kernel k and the corresponding norm is denoted by $\|\bullet\|_{\mathcal{H}_k}$. In this paper, we use MMD due to its computational simplicity and popularity in machine learning community.

Additional notation: For generic sequences $\{\varpi(N)\}$ and $\{\theta(N)\}$, the notation $\varpi(N) \gtrsim \theta(N)$ (resp. $\varpi(N) \lesssim \theta(N)$) means that there exists a sufficiently large constant (resp. small) constant $c_1 > 0$ (resp. $c_2 > 0$) such that $\varpi(N) \geq c_1 \theta(N)$ (resp. $\varpi(N) \leq c_2 \theta(N)$). We use $\varpi(N) \asymp \theta(N)$ when $\varpi(N) \gtrsim \theta(N)$ and $\varpi(N) \lesssim \theta(N)$. For matrix and vector norms, we use $\|\bullet\|_{\ell_q}$ to denote either the vector ℓ_q -norm or operator norm induced by the vector ℓ_q -norm, for $1 \leq q < \infty$, when there is no confusion. For any random variable X , we use $L_{\mathbb{P}}^q(X)$ to denote the class of all measurable functions with finite q -th moments for $1 \leq q \leq \infty$, where the underlying probability measure is $\mathbb{P}$. We may also write it as $L_{\mathbb{P}}^q$ when there is no confusion about the underlying random vectors. Then the L^q -norm is denoted by $\|\bullet\|_{L_{\mathbb{P}}^q(X)}$. We use $\|\bullet\|_{\infty}$ to denote the point-wise sup-norm for a vector-valued function. For example, when π is a deterministic policy, $\|\pi\|_{\infty}$ returns a vector with sup-norm on each coordinate for $\pi \in \Pi$. In addition, we often use (S, A, R, S') or (S, A, S') to represent a generic transition tuple, $Z = (S, A)$ and $\mathcal{Z} = \mathcal{S} \times \mathcal{A}$.

3 Policy Learning in Batch RL

3.1 Warm up: Contextual Bandits under Singularity

We start with an offline contextual bandit problem (Manski, 2004) as an example to illustrate the challenge of policy learning under singularity and our main idea to address it. Note that the contextual bandit problem is a special case of batch RL, in with $T = 1$, $\gamma = 0$, the visitation probability measure is $d_0^\pi = \nu \times \pi$, and the policy value function is $\mathcal{V}(\pi) = \mathbb{E}_{S_0 \sim \nu} [R_0] = \mathbb{E}_{(S_0, A_0) \sim \nu \times \pi} [r(S_0, A_0)] = \mathbb{E}_{S_0 \sim \nu} [r(S_0, \pi(S_0))]$.

Suppose we observe a batch dataset $\mathcal{D}_N = \{S_{i,0}, A_{i,0}, R_{i,0}\}_{1 \leq i \leq N}$ that contains i.i.d. copies from (S_0, A_0, R_0) generated by a DGP $d_0^{\pi^b} = \nu_0 \times \pi^b$, with a behavior policy π^b , and $S_0 \sim \nu_0$ for an unknown distribution ν_0 . Our goal is to find an in-class optimal policy

$$\pi^* \in \arg \max_{\pi \in \Pi} \left\{ \mathbb{E}_{(S_0, A_0) \sim \nu \times \pi} [r(S_0, A_0)] \right\}, \quad (11)$$

without imposing $\nu \times \pi^* \ll \nu_0 \times \pi^b$ or a parametric functional form of the reward function $r(\cdot, \cdot)$. In particular, we want to allow for both $\pi^* \perp \pi^b$ (e.g., deterministic optimal policy vs continuous behavior policy), and $\nu \perp \nu_0$ (covariates shifts). Most existing methods for solving (11) rely on a key assumption that $\nu \times \pi \ll \nu_0 \times \pi^b$ for all $\pi \in \Pi$ (e.g., Qian and Murphy, 2011; Bhattacharya and Dupas, 2012; Zhao et al., 2012). Exceptions are Chen et al. (2016), which adopted the kernel smoothing techniques to relax $\pi \ll \pi^b$, and Chernozhukov et al. (2019), which imposed some (parametric) structural assumption on r . However, these methods cannot handle general settings such as that ν is also singular to ν_0 , i.e., there exists a *covariate shift* problem.

We now briefly describe our method for policy learning without requiring absolute continuity of any candidate distribution (i.e., $\nu \times \pi$) with respect to the data distribution ($\nu_0 \times \pi^b$) in this contextual bandit problem. For any given policy $\pi \in \Pi$, let $\tilde{r}(S_0, \pi(S_0))$ be any estimator of the reward function $r(S_0, \pi(S_0))$, and then $\tilde{\mathcal{V}}(\pi) = \mathbb{E}_{S_0 \sim \nu} [\tilde{r}(S_0, \pi(S_0))]$

is an estimate of $\mathcal{V}(\pi)$. We have: $\mathcal{V}(\pi) - \tilde{\mathcal{V}}(\pi) = \mathbb{E}_{(S_0, A_0) \sim \nu \times \pi} [(r - \tilde{r})(S_0, A_0)]$. Applying the Lebesgue Decomposition (8), we have

$$\begin{aligned} \mathcal{V}(\pi) - \tilde{\mathcal{V}}(\pi) = & \underbrace{\tilde{\lambda}_1^\pi(\mathcal{Z}) \times \mathbb{E}_{(S_0, A_0) \sim \nu_0 \times \pi^b} \left[\frac{\lambda_1^\pi(S_0, A_0)}{(\nu_0 \times \pi^b)(S_0, A_0)} (r - \tilde{r})(S_0, A_0) \right]}_{\text{absolutely continuous part}} \\ & + \underbrace{\tilde{\lambda}_2^\pi(\mathcal{Z}) \times \mathbb{E}_{(S_0, A_0) \sim \lambda_2^\pi} [(r - \tilde{r})(S_0, A_0)]}_{\text{singular part}}. \end{aligned}$$

In order to have an accurate estimation of $\mathcal{V}(\pi)$, one needs to find an $\tilde{r}$ such that the absolutely continuous and singular parts in the above equality are minimized. For the absolutely continuous part, since λ_1^π is unknown, define $\mathcal{W}$ as some class of symmetric functions and assume $\lambda_1^\pi/(\nu_0 \times \pi^b) \in \mathcal{W}$, we can show that

$$\left| \mathbb{E}_{(S_0, A_0) \sim \nu_0 \times \pi^b} \left[\frac{\lambda_1^\pi(S_0, A_0)}{(\nu_0 \times \pi^b)(S_0, A_0)} (r - \tilde{r})(S_0, A_0) \right] \right| \leq \sup_{w \in \mathcal{W}} \mathbb{E}[w(S_0, A_0)(R_0 - \tilde{r}(S_0, A_0))], \quad (12)$$

where the right-hand side of the above inequality can be properly controlled using the batch data because (S_0, A_0) now follows $d_0^{\pi^b} = \nu_0 \times \pi^b$. To handle the singular part, we adopt the idea of distributionally robust optimization (e.g., a review paper by [Rahimian and Mehrotra, 2019](#)) with IPM τ_Ξ , such as TV, MMD and W_1 distances. If $r - \tilde{r} \in \Xi$ then

$$|\mathbb{E}_{(S_0, A_0) \sim \lambda_2^\pi} [(r - \tilde{r})(S_0, A_0)]| \leq |\mathbb{E}_{(S_0, A_0) \sim \nu_0 \times \pi^b} [(r - \tilde{r})(S_0, A_0)]| + \tau_\Xi(\lambda_2^\pi, \nu_0 \times \pi^b) \|r - \tilde{r}\|_\Xi. \quad (13)$$

Combining (12) and (13) with $\{\pm 1\} \in \mathcal{W}$, we obtain

$$|\mathcal{V}(\pi) - \tilde{\mathcal{V}}(\pi)| \leq \sup_{w \in \mathcal{W}} \mathbb{E}[w(S_0, A_0)(R_0 - \tilde{r}(S_0, A_0))] + \tilde{\lambda}_2^\pi(\mathcal{Z}) \tau_\Xi(\lambda_2^\pi, \nu_0 \times \pi^b) \|r - \tilde{r}\|_\Xi.$$

It is easy to see that if $\tilde{r} = r$, the right-hand side (RHS) of the above inequality vanishes. This motivates us to estimate the reward function r and then $\mathcal{V}(\pi)$ by minimizing the above RHS expression. Once we have a good estimate for $\mathcal{V}(\pi)$, an optimization routine can be

implemented to estimate π^* , where we incorporate the idea of pessimism as follows:

$$\begin{aligned}
& \max_{\pi \in \Pi} \min_{\tilde{r} \in \mathcal{R}} \mathbb{E}_{S_0 \sim \nu} [\tilde{r}(S_0, \pi(S_0))], \\
& \text{subject to} \quad \sup_{w \in \mathcal{W}} \bar{\mathbb{E}}_N [w(S_0, A_0)(R_0 - \tilde{r}(S_0, A_0))] \leq \varepsilon_1 \\
& \quad \|\hat{\mathbb{E}}[R_0 - \tilde{r} | S_0 = \bullet, A_0 = \bullet]\|_{\Xi} \leq \varepsilon_2,
\end{aligned} \tag{14}$$

where $\mathcal{R}$ is some pre-specified class of smooth functions for modeling the true reward function r , $\bar{\mathbb{E}}_N$ is the empirical average over the batch data $\mathcal{D}_N$, and $\hat{\mathbb{E}}[R_0 - \tilde{r} | S_0 = \bullet, A_0 = \bullet]$ is a consistent estimate, say a kernel ridge regression in our paper, of $\mathbb{E}[R_0 - \tilde{r} | S_0 = \bullet, A_0 = \bullet]$. Two constants $\varepsilon_1 > 0$ and $\varepsilon_2 > 0$ are used to quantify the uncertainty for estimating r and also control the degree of pessimism. The problem (14) can be easily solved by its dual program. See Section 5 and Supplementary Material for more details.

3.2 OPE with Potential Singularity in RL

We now consider the sequential decision-making setting, i.e., $T > 1$. Recall that for any given policy π , the target of OPE is to use the batch data to estimate the policy value $\mathcal{V}(\pi)$ defined in (1) or equivalently (4) or (6). A direct approach is via estimating Q^π defined in (3). Let $\mathcal{F}$ be some pre-specified nonparametric class of smooth functions over $\mathcal{Z}$, which is used to model Q^π with $\pi \in \Pi$. Let $\mathcal{B}^\pi : \mathcal{F} \mapsto \mathcal{F}$ be the Bellman operator: $\mathcal{B}^\pi \tilde{Q}(\bullet, \bullet) = \mathbb{E}[R + \gamma \tilde{Q}(S', \pi(S')) | S = \bullet, A = \bullet]$ for any transition tuple (S, A, R, S') . Define $\mathcal{T}^\pi \tilde{Q} := \mathcal{B}^\pi \tilde{Q} - \tilde{Q}$ as the Bellman residual operator. Then $\mathcal{T}^\pi Q^\pi = 0$ by the definition of Q^π . Let $\tilde{Q}$ be any estimator for Q^π and define the corresponding estimator for the policy value as $\tilde{\mathcal{V}}(\pi) = (1 - \gamma) \mathbb{E}_{S_0 \sim \nu} [\tilde{Q}(S_0, \pi(S_0))]$. Then we have the following lemma to characterize the OPE error of $\tilde{\mathcal{V}}(\pi)$ to $\mathcal{V}(\pi)$:

Lemma 1 *Under Assumptions 1 and 2, for any $\pi \in \Pi$ and $\tilde{Q} \in \mathcal{F}$, we have*

$$\mathcal{V}(\pi) - \tilde{\mathcal{V}}(\pi) = \mathbb{E}_{(S,A) \sim d_\gamma^\pi} [R + \gamma \tilde{Q}(S', \pi(S')) - \tilde{Q}(S, A)] = \mathbb{E}_{(S,A) \sim d_\gamma^\pi} [(\mathcal{T}^\pi \tilde{Q})(S, A)]. \tag{15}$$

As mentioned before, we consider the most challenging setting that $d_\gamma^\pi \ll d_\gamma^{\pi^b}$ fails to hold and singularity arises. By the Lebesgue Decomposition (8), for any $\pi \in \Pi$, the OPE error can be decomposed into

$$\mathcal{V}(\pi) - \tilde{\mathcal{V}}(\pi) = \underbrace{\tilde{\lambda}_1^\pi(\mathcal{Z}) \mathbb{E}_{(S,A) \sim \lambda_1^\pi} \left[(\mathcal{T}^\pi \tilde{Q})(S, A) \right]}_{\text{absolute continuous part}} + \underbrace{\tilde{\lambda}_2^\pi(\mathcal{Z}) \mathbb{E}_{(S,A) \sim \lambda_2^\pi} \left[(\mathcal{T}^\pi \tilde{Q})(S, A) \right]}_{\text{singular part}}.$$

For the absolutely continuous part of λ_1^π with respect to $d_\gamma^{\pi^b}$, we introduce the Radon-Nikodym derivative $\omega^\pi(s, a) = \frac{\lambda_1^\pi(s, a)}{d_\gamma^{\pi^b}(s, a)}$ for every $(s, a) \in \mathcal{Z}$ and apply the change of measure. For the difficult singularity part of λ_2^π with respect to $d_\gamma^{\pi^b}$ we apply the distributionally robust approach with IPM $\tau_\Xi(\lambda_2^\pi, d_\gamma^{\pi^b})$ and obtain that

$$\begin{aligned} \mathcal{V}(\pi) - \tilde{\mathcal{V}}(\pi) &= \tilde{\lambda}_1^\pi(\mathcal{Z}) \bar{\mathbb{E}} \left[w^\pi(S, A) (\mathcal{T}^\pi \tilde{Q})(S, A) \right] + \tilde{\lambda}_2^\pi(\mathcal{Z}) \bar{\mathbb{E}} \left[(\mathcal{T}^\pi \tilde{Q})(S, A) \right] \\ &\quad + \tilde{\lambda}_2^\pi(\mathcal{Z}) \left(\mathbb{E}_{(S,A) \sim \lambda_2^\pi} \left[(\mathcal{T}^\pi \tilde{Q})(S, A) \right] - \bar{\mathbb{E}} \left[(\mathcal{T}^\pi \tilde{Q})(S, A) \right] \right). \end{aligned} \quad (16)$$

We can now present our key lemma to bound the OPE error of $\tilde{\mathcal{V}}(\pi)$.

Lemma 2 *Let Assumptions 1 and 2 hold, $\mathcal{W}$ be a symmetric class of real-valued functions defined over $\mathcal{Z}$ and $\{\pm 1\} \in \mathcal{W}$. For any $\pi \in \Pi$ and $\tilde{Q} \in \mathcal{F}$ such that $\mathcal{T}^\pi \tilde{Q} \in \Xi$, we have:*

$$|\mathcal{V}(\pi) - \tilde{\mathcal{V}}(\pi)| \leq \sup_{w \in \mathcal{W}} \bar{\mathbb{E}} \left[w(S, A) \left(R + \gamma \tilde{Q}(S', \pi(S')) - \tilde{Q}(S, A) \right) \right] + \tilde{\lambda}_2^\pi(\mathcal{Z}) \tau_\Xi(d_\gamma^{\pi^b}, \lambda_2^\pi) \|\mathcal{T}^\pi \tilde{Q}\|_\Xi. \quad (17)$$

(i) *If MMD is used, let the kernel $k(\bullet, \bullet) : \mathcal{Z} \times \mathcal{Z} \rightarrow \mathbb{R}$ be measurable with respect to λ_2^π and $d_\gamma^{\pi^b}$, and $\max\{\bar{\mathbb{E}}[\sqrt{k(Z, Z)}], \mathbb{E}_{Z \sim \lambda_2^\pi}[\sqrt{k(Z, Z)}]\} < \infty$, then:*

$$|\mathcal{V}(\pi) - \tilde{\mathcal{V}}(\pi)| \leq \sup_{w \in \mathcal{W}} \bar{\mathbb{E}}[w(S, A)(R + \gamma \tilde{Q}(S', \pi(S')) - \tilde{Q}(S, A))] + \tilde{\lambda}_2^\pi(\mathcal{Z}) \text{MMD}_k(d_\gamma^{\pi^b}, \lambda_2^\pi) \|\mathcal{T}^\pi \tilde{Q}\|_{\mathcal{H}_k}. \quad (18)$$

(ii) *If TV is used, then $\tau_\Xi(d_\gamma^{\pi^b}, \lambda_2^\pi) \|\mathcal{T}^\pi \tilde{Q}\|_\Xi = \text{TV}(d_\gamma^{\pi^b}, \lambda_2^\pi) \|\mathcal{T}^\pi \tilde{Q}\|_\infty$; (iii) *If W_1 is used, then $\tau_\Xi(d_\gamma^{\pi^b}, \lambda_2^\pi) \|\mathcal{T}^\pi \tilde{Q}\|_\Xi = W_1(d_\gamma^{\pi^b}, \lambda_2^\pi) \|\mathcal{T}^\pi \tilde{Q}\|_{\text{Lip}}$, where $\|\bullet\|_{\text{Lip}}$ is the Lipschitz norm.**

insight and trade-off; singularity times smoothness for extrapolation Lemma 2 provides an upper bound for the estimation error of OPE using any estimator $\tilde{Q}$ for Q^π under various IPMs. It is easy to see that, by Bellman equation, the right-hand-side (RHS) of (17) is

zero if $\tilde{Q} = Q^\pi$, which demonstrates the validity of using it as an objective function to find a good estimate for Q^π . Since the first term on the RHS of (17) can be calibrated by the batch data distribution, it can be estimated efficiently under mild conditions. The second term on the RHS of (17) is more difficult to estimate, whose convergence rate is typically slower than the first term. For computational convenience, we will focus on MMD hence (18) to illustrate our new algorithms in the remainder of this paper. Nevertheless, under some conditions there are no substantial differences for these IPM metrics. For example, see Vayer and Gribonval (2023) for the equivalence between MMD and Wasserstein metrics. Based on the above analysis, one can estimate Q^π by solving the following minimax problem and perform OPE.

$$\inf_{\tilde{Q} \in \mathcal{F}} \left[\sup_{w \in \mathcal{W}} \mathbb{E} \left[w(S, A) \left(R + \gamma \tilde{Q}(S', \pi(S')) - \tilde{Q}(S, A) \right) \right] + \tilde{\lambda}_2^\pi(\mathcal{Z}) \text{MMD}_k(d_\gamma^{\pi^b}, \lambda_2^\pi) \|\mathcal{T}^\pi \tilde{Q}\|_{\mathcal{H}_k} \right]. \quad (19)$$

This new method will serve as the foundation of our proposed STEEL algorithm below.

Make an algorithm section

3.3 Policy Optimization under Potential Singularity in RL

Recall that our goal is to leverage the batch data $\mathcal{D}_N$ to estimate the in-class optimal policy π^* defined in (2). For any policy π , we can implement an empirical version of (19) to obtain an estimate of Q^π and then perform the policy evaluation. This motivates us to develop a pessimistic policy iteration algorithm for finding π^* . Pessimism in batch RL serves as the main tool for efficient policy optimization by quantifying the uncertainty of the estimation (e.g., Kumar et al., 2019; Bai et al., 2022).

A key step in our proposed algorithm is the construction of the following two feasible uncertainty sets for $Q^\pi \in \mathcal{F}$:

$$\begin{aligned} \Omega_1(\pi, \mathcal{F}, \mathcal{W}, \varepsilon_{NT}^{(1)}) &= \{Q \in \mathcal{F} \mid \sup_{w \in \mathcal{W}} \bar{\mathbb{E}}_{NT} [w(S, A)(R + \gamma Q(S', \pi(S')) - Q(S, A))] \leq \varepsilon_{NT}^{(1)}\}, \\ \Omega_2(\pi, \mathcal{F}, \varepsilon_{NT}^{(2)}) &= \left\{ Q \in \mathcal{F} \mid \|\hat{\mathcal{T}}^\pi Q\|_{\mathcal{H}_k} \leq \varepsilon_{NT}^{(2)} \right\}, \end{aligned} \quad (20)$$

where $\varepsilon_{NT}^{(1)} > 0$ and $\varepsilon_{NT}^{(2)} > 0$ are some tuning parameters possibly depending on N and T , which will be specified later. Here $\bar{\mathbb{E}}_{NT}[f(S, A, R, S')] = \frac{1}{NT} \sum_{i=1}^N \sum_{t=0}^T f(S_{i,t}, A_{i,t}, R_{i,t}, S_{i,t+1})$ for any function f , and $\hat{\mathcal{T}}^\pi$ is an estimator of $\mathcal{T}^\pi$ to be specified later. We propose to estimate the in-class optimal policy $\pi^* \in \Pi$ via

$$\hat{\pi} \in \arg \max_{\pi \in \Pi} \min_{Q \in \Omega_1(\pi, \mathcal{F}, \mathcal{W}, \varepsilon_{NT}^{(1)}) \cap \Omega_2(\pi, \mathcal{F}, \varepsilon_{NT}^{(2)})} (1 - \gamma) \mathbb{E}_{S_0 \sim \nu} [Q(S_0, \pi(S_0))], \quad (21)$$

i.e., maximizing the worst-case policy value within the uncertainty sets (the idea of Pessimism). Note that, compared with OPE using (19), our optimization problem (21) is feasible, *free* of $\tilde{\lambda}_2^\pi(\mathcal{Z})$ and $\text{MMD}_k(d_\gamma^{\pi^b}, \lambda_2^\pi)$. Theorem 2 in Section 4 establishes a regret guarantee for $\hat{\pi}$ in finding π^* under minimal data coverage assumption.

3.4 Dual Problem and Algorithm

The policy optimization problem (21), in its original form, may be difficult to solve because of the constraint set for Q , especially when $\mathcal{F}$ and $\mathcal{W}$ are highly complex such as neural networks. We now propose to solve (21) via its dual formulation.

Let $\rho = (\rho_1, \rho_2)$ be dual variables. Define a Lagrangian function as

$$\begin{aligned} L_{NT}(Q, \rho, \pi) := & (1 - \gamma) \mathbb{E}_{S_0 \sim \nu} [Q(S_0, \pi(S_0))] + \rho_2 \times \left\{ \|\hat{\mathcal{T}}^\pi Q\|_{\mathcal{H}_k} - \varepsilon_{NT}^{(2)} \right\} \\ & + \rho_1 \times \left\{ \sup_{w \in \mathcal{W}} \bar{\mathbb{E}}_{NT} [w(S, A) (R + \gamma Q(S', \pi(S')) - Q(S, A))] - \varepsilon_{NT}^{(1)} \right\}. \end{aligned} \quad (22)$$

We can estimate the optimal policy π^* by solving the computationally easier dual problem:

$$\hat{\pi}_{\text{dual}} \in \arg \max_{\pi \in \Pi} \max_{\rho \succeq 0} \min_{Q \in \mathcal{F}} L_{NT}(Q, \rho, \pi), \quad (23)$$

where $\succeq$ refers to the component-wise comparison.

By the weak duality, $\max_{\rho \succeq 0} \min_{Q \in \mathcal{F}} L_{NT}(Q, \rho, \pi) \leq \min_{Q \in \Omega_1(\pi, \mathcal{F}, \mathcal{W}, \varepsilon_{NT}^{(1)}) \cap \Omega_2(\pi, \mathcal{F}, \varepsilon_{NT}^{(2)})} (1 - \gamma) \mathbb{E}_{S_0 \sim \nu} [Q(S_0, \pi(S_0))]$. Therefore, solving Problem (23) can also be viewed as a policy optimization algorithm with pessimism. However, the strong duality for the primal and

dual problems may not hold if the primal problem (21) is not convex with respect to $Q \in \mathcal{F}$. In general, the existence of the duality gap will result in a non-negligible regret of $\hat{\pi}_{\text{dual}}$ in finding π^* . Nevertheless, in Section 4 below, we show that under one additional mild assumption — $\mathcal{F}$ being a convex class, $\hat{\pi}_{\text{dual}}$ indeed can achieve the same regret guarantee as that of $\hat{\pi}$. See Algorithm 1 and for implementation details, we refer to Section 2 of the appendix. [algorithm more details](#)

Algorithm 1: A pseudo algorithm of STEEL.

Input: Data $\mathcal{D}_N$, kernel function $k(\cdot, \cdot)$, discount factor γ , initial state distribution

ν , function class $\mathcal{F}$ and Π , tuning parameters ζ_{NT} , $\varepsilon_{NT}^{(1)}$, and $\varepsilon_{NT}^{(2)}$.

2 Initialize the values of $\rho = (\rho_1, \rho_2)$, Q , and π

3 Compute the pairwise kernel matrix

$$K \in \mathbb{R}^{NT \times NT} \quad \text{with} \quad K_{i,j} = k(Z_{\lfloor i/T \rfloor + 1, i \bmod T - 1}, Z_{\lfloor j/T \rfloor + 1, j \bmod T - 1}).$$

4 **repeat**

5 Take a few gradient descent steps with respect to Q with learning rate lr_Q .

6 Take gradient ascent steps with respect to ρ_1 , ρ_2 and π with some specified learning rates.

7 **until** *convergence*;

Result: An estimated Policy $\hat{\pi}$

3.5 Adaptive STEEL

[Explain the details of the algorithm](#) In this subsection, we propose an adaptive version of STEEL to finding the optimal policy under singularity. It consists of three steps. In the first step, we run STEEL by only considering the feasible constraint set $\Omega_2(\pi, \mathcal{F}, \varepsilon_{NT}^{(2)})$ with a sufficiently large $\varepsilon_{NT}^{(2)}$. Essentially, we implement the algorithm without assuming existence of any overlapping region between the behavior policy and the optimal one. Denote the

resulting estimated policy as $\hat{\pi}_0$ and the estimated Q -function as $\hat{Q}$ after solving (21) or (23). In the second step, based on $\hat{\pi}_0$, we aim to construct a single feasible constraint set Ω that takes account for both singular and absolutely continuous parts. Basically, we leverage the initially estimated policy $\hat{\pi}_0$ to calibrate the absolutely continuous and singular parts with respect to the batch data distribution. To achieve this goal, we need to estimate several related quantities including $\tilde{\lambda}_1^{\hat{\pi}_0}, \tilde{\lambda}_2^{\hat{\pi}_0}, \lambda_1^{\hat{\pi}_0}, \lambda_2^{\hat{\pi}_0}, \delta^{\hat{\pi}_0} = \text{MMD}(\lambda_2^{\hat{\pi}_0}, d_{\gamma}^{\pi^b})$, and $\omega^{\hat{\pi}_0}$. One can first estimate the behavior policy and transition probability using the batch data. Then $\bar{d}_T^{\pi^b}$ and $d_{\gamma}^{\hat{\pi}_0}$ can be estimated correspondingly, where the estimators are denoted by $\hat{d}_T^{\pi^b}$ and $\hat{d}_{\gamma}^{\hat{\pi}_0}$ respectively. To find $\tilde{\lambda}_1^{\hat{\pi}_0} \times \omega^{\hat{\pi}_0}$, one can solve the following minimization problem: $\underset{\zeta \in \mathcal{W}}{\text{minimize}} \sup_{f \in \mathcal{H}_k} \left| \mathbb{E}_{(S,A) \sim d_{\gamma}^{\hat{\pi}_0}} [f(S, A)] - \mathbb{E}[\zeta(S, A)f(S, A)] \right|$. Denote the optimal solution as $\hat{\zeta}^{\hat{\pi}_0}$. Based on the relationship that $\tilde{\lambda}_1^{\hat{\pi}_0} \times \lambda_1^{\hat{\pi}_0} = \hat{\zeta}^{\hat{\pi}_0} \times d_{\gamma}^{\pi^b}$, we can obtain an estimator of $\tilde{\lambda}_1^{\hat{\pi}_0}$ denoted by $\widehat{\tilde{\lambda}}_1^{\hat{\pi}_0} = \int_{S \times \mathcal{A}} \hat{\zeta}^{\hat{\pi}_0}(s, a) \times \hat{d}_T^{\pi^b}(s, a) ds da$ and an estimator of $\lambda_1^{\hat{\pi}_0}(s, a)$ denoted by $\hat{\lambda}_1^{\hat{\pi}_0}(s, a) = \hat{\zeta}^{\hat{\pi}_0}(s, a) \times d_{\gamma}^{\pi^b}(s, a) / \widehat{\tilde{\lambda}}_1^{\hat{\pi}_0}$. By the Lebesgue's Decomposition, we can find $\tilde{\lambda}_2^{\hat{\pi}_0} \times \lambda_2^{\hat{\pi}_0} = d_{\gamma}^{\hat{\pi}_0} - \hat{\zeta}^{\hat{\pi}_0} \times d_{\gamma}^{\pi^b}$. Correspondingly, we can construct their estimators denoted by $\widehat{\tilde{\lambda}}_2^{\hat{\pi}_0}$ and $\hat{\lambda}_2^{\hat{\pi}_0}$ correspondingly. Finally, an estimator of $\delta^{\hat{\pi}_0}$ denoted by $\hat{\delta}^{\hat{\pi}_0}$ can be computed by standard methods such as (Gretton et al., 2012). Based on the aforementioned quantities, we construct the following uncertainty set denoted by Ω .

$$\begin{aligned} \Omega = \{ & Q \in \mathcal{F} \mid \widehat{\tilde{\lambda}}_1^{\hat{\pi}_0}(\mathcal{Z}) \times \mathbb{E}_{(S,A) \sim \hat{\lambda}_1^{\hat{\pi}_0}} [R + \gamma Q(S', \pi(S')) - Q(S, A)] \\ & + \widehat{\tilde{\lambda}}_2^{\hat{\pi}_0}(\mathcal{Z}) \times \mathbb{E}_{NT} [R + \gamma Q(S', \pi(S')) - Q(S, A)] + \widehat{\tilde{\lambda}}_2^{\hat{\pi}_0}(\mathcal{Z}) \times \hat{\delta}^{\hat{\pi}_0} \|\hat{\mathcal{T}}^{\pi} Q\|_{\mathcal{H}_k} \leq \varepsilon_0 \}, \end{aligned} \quad (24)$$

where

$$\begin{aligned} \varepsilon_0 = & \widehat{\tilde{\lambda}}_1^{\hat{\pi}_0}(\mathcal{Z}) \times \mathbb{E}_{(S,A) \sim \hat{\lambda}_1^{\hat{\pi}_0}} [R + \gamma \hat{Q}(S', \hat{\pi}_0(S')) - \hat{Q}(S, A)] \\ & + \widehat{\tilde{\lambda}}_2^{\hat{\pi}_0}(\mathcal{Z}) \times \mathbb{E}_{NT} [R + \gamma \hat{Q}(S', \hat{\pi}_0(S')) - \hat{Q}(S, A)] + \widehat{\tilde{\lambda}}_2^{\hat{\pi}_0}(\mathcal{Z}) \times \hat{\delta}^{\hat{\pi}_0} \|\hat{\mathcal{T}}^{\hat{\pi}_0} \hat{Q}\|_{\mathcal{H}_k}. \end{aligned} \quad (25)$$

Note that Ω does not rely on any tuning parameter. The last step is to run our proposed STEEL again with the single constraint set Ω , which gives an adaptive policy $\hat{\pi}_1$. See

Algorithm 2 for the full adaptive STEEL.

Algorithm 2: A pseudo algorithm of adaptive STEEL.

Input: Data $\mathcal{D}_N = \{(S_{i,t}, A_{i,t}, R_{i,t}, S_{i,t+1})\}_{0 \leq t < T, 1 \leq i \leq N}$

2 Run STEEL algorithm with the constraint set Ω_2 only and a sufficiently large $\varepsilon_{NT}^{(2)}$.

Output $\hat{\pi}_0$ and $\hat{Q}$.

3 Estimate $\tilde{\lambda}_1^{\hat{\pi}_0}, \tilde{\lambda}_2^{\hat{\pi}_0}, \lambda_1^{\hat{\pi}_0}, \lambda_2^{\hat{\pi}_0}, \delta^{\hat{\pi}_0}, \bar{d}_T^{\pi^b}$, and $\omega^{\hat{\pi}_0}$, and compute ε_0 and construct Ω given in Equations (25) of the appendix and (24) respectively.

4 Run STEEL with the constraint set Ω .

Result: An estimated policy $\hat{\pi}_1$

4 Theoretical Results

delete obvious results and show important results

The performance of a policy optimization algorithm is often measured by the difference between the value of the (in-class) optimal policy and that of the estimated one, which is referred to as the *regret*. The regret of an estimated policy $\tilde{\pi} \in \Pi$ is defined as

$$\text{Regret}(\tilde{\pi}) := \mathcal{V}(\pi^*) - \mathcal{V}(\tilde{\pi}) \geq 0. \quad (26)$$

In this section, we derive the finite-sample upper bounds for both $\text{Regret}(\hat{\pi})$ and $\text{Regret}(\hat{\pi}_{\text{dual}})$.

We leave the theoretical analysis of adaptive STEEL as future work.

4.1 Technical Assumptions

We first present several important assumptions and their implications. Additional regularity conditions are postponed to Appendix 3.

Assumption 4 *The transition-tuple process $\{(S_t, A_t, S_{t+1})\}_{t \geq 0}$ induced by π^b is stationary and exponentially β -mixing. The corresponding β -mixing coefficient at time lag j satisfies that $\beta(j) \leq \beta_0 \exp(-\beta_1 j)$ for $\beta_0 \geq 0$ and $\beta_1 > 0$.*

Denote the induced stationary distribution of (S_t, A_t) as d^{π^b} . Our batch data consist of multiple trajectories, where observations in each trajectory follow a MDP and thus are dependent. Assumption 4 characterizes the dependency among the transition tuples that appear in the Bellman residual and kernel-regression arguments, which has been used in recent works (e.g., Shi, Zhang, Lu and Song, 2020; Chen and Qi, 2022). See Bradley (2005) for definition of the exponentially β -mixing.

In the following we define a regularized Bellman residual operator $\mathcal{T}_{\zeta_{NT}}^{\pi} : \mathcal{F} \rightarrow \mathcal{H}_k$ as

$$\mathcal{T}_{\zeta_{NT}}^{\pi} Q := \arg \min_{f \in \mathcal{H}_k} \mathbb{E} \left[(R + \gamma Q(S', \pi(S')) - Q(S, A) - f(S, A))^2 \right] + \zeta_{NT} \|f\|_{\mathcal{H}_k}^2, \quad (27)$$

for every $Q \in \mathcal{F}$. Consider the following class of vector-valued functions as

$$\begin{aligned} \mathcal{G} = \{ & g : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathcal{H}_k \mid g(s, a, s') = (Y(Q, \pi, s, a, s') - \mathcal{T}_{\zeta_{NT}}^{\pi} Q(s, a)) k((s, a), \bullet) \\ & \text{with } Q \in \mathcal{F}, \pi \in \Pi \}, \end{aligned}$$

where $Y(Q, \pi, s, a, s') = \tilde{R}(s, a, s') + \gamma Q(s', \pi(s')) - Q(s, a)$. For $g \in \mathcal{G}$, we use $\|g\|_{\mathcal{H}_{k,\infty}}$ to denote $\sup_{(s,a,s') \in \mathcal{S} \times \mathcal{A} \times \mathcal{S}} \|g(s, a, s')\|_{\mathcal{H}_k}$. In the following, we quantify the complexities of function classes via the ϵ -covering number. An ϵ -covering number of a set Θ , denoted by $\mathcal{N}(\epsilon, \Theta, \tilde{d})$, is the infimum of the cardinality of ϵ -cover of Θ measured by a (semi)-metric $\tilde{d}$.

Assumption 5 *The following conditions hold:*

(a) *The weighting class $\mathcal{W}$ is symmetric, $\{\pm 1\} \subseteq \mathcal{W}$, and $\sup_{w \in \mathcal{W}} \|w\|_{\infty} \leq c_{\mathcal{W}} < \infty$. In addition, $\sup_{Q \in \mathcal{F}} \|Q\|_{\infty} \leq c_{\mathcal{F}} < \infty$. For any $Q \in \mathcal{F}$, $s \in \mathcal{S}$, and $a_1, a_2 \in \mathcal{A}$,*

$$|Q(s, a_1) - Q(s, a_2)| \lesssim \|a_1 - a_2\|_{\ell_1}. \quad (28)$$

In addition, there is a finite constant $v(\mathcal{F}) > 0$ such that for any $\epsilon > 0$,

$$\mathcal{N}(\epsilon, \mathcal{W}, \|\bullet\|_{\infty}) \lesssim \mathcal{N}(\epsilon, \mathcal{F}, \|\bullet\|_{\infty}) \lesssim \left(\frac{1}{\epsilon}\right)^{v(\mathcal{F})}. \quad (29)$$

(b) Π is a class of deterministic policies (i.e., $\pi : \mathcal{S} \rightarrow \mathcal{A}$). The action space $\mathcal{A}$ is bounded.

There are finite constants $v_i(\Pi) > 0$, $1 \leq i \leq d_A$, such that for any $\epsilon > 0$,

$$\mathcal{N}(\epsilon, \Pi_i, \|\bullet\|_\infty) \lesssim \left(\frac{1}{\epsilon}\right)^{v_i(\Pi)}. \quad (30)$$

Let $v(\Pi) = \sum_{i=1}^{d_A} v_i(\Pi)$.

(c) $\sup_{g \in \mathcal{G}} \|g\|_{\mathcal{H}_{k,\infty}} \leq c_{\mathcal{G}} < \infty$. There is a finite constant $v(\mathcal{G}) > 0$ such that for any $\epsilon > 0$,

$$\mathcal{N}(\epsilon, \mathcal{G}, \|\bullet\|_{\mathcal{H}_{k,\infty}}) \lesssim \left(\frac{1}{\epsilon}\right)^{v(\mathcal{G})}. \quad (31)$$

Assumption 5 imposes metric entropy conditions on function classes $\mathcal{F}$, $\mathcal{W}$, Π and $\mathcal{G}$. For simplicity, we consider uniformly bounded classes for deriving the exponential inequalities for both scalar-value and vector-value function classes (Van Der Vaart and Wellner, 1996; Park and Muandet, 2022). In Assumption 5 (b), we focus on the policy class that consists of deterministic policies because there always exists an optimal policy that is deterministic. We remark that $v(\mathcal{F})$, $v(\Pi)$ and $v(\mathcal{G})$ could increase with respect to N and T . In practice, one can specify $\mathcal{F}$, Π and $\mathcal{W}$ (including complicated function classes such as neural networks) and compute their corresponding complexities. In Appendix 3 we provide a sufficient condition for computing $v(\mathcal{G})$. The Lipschitz-typed condition in (28) is imposed so that the ϵ -covering number of the following class of functions:

$$\tilde{\mathcal{F}} := \{(S, A, S') \rightarrow R + \gamma Q(S', \pi(S')) - Q(S, A) \mid Q \in \mathcal{F}, \pi \in \Pi\}$$

could be properly controlled by the ϵ -covering numbers of Π_i for $1 \leq i \leq d_A$ and $\mathcal{F}$ (Van Der Vaart and Wellner, 1996).

Next, let $c \in (1/2, 3/2]$ be a finite constant, $\{e_j\}_{j \geq 1}$ be a non-increasing sequence of eigenvalues of the convolution operator defined in Appendix 3 towards 0, and $\{\phi_j\}_{j \geq 1}$ be

orthonormal bases of $L^2_{\bar{d}_T^\pi}(\mathcal{Z})$. Define a Hilbert space $\mathcal{H}_k^{2c}$ as

$$\mathcal{H}_k^{2c} := \left\{ f = \sum_{i=1}^{\infty} \bar{e}_i \phi_i \mid \forall i \geq 1, \bar{e}_i \in \mathbb{R} \text{ such that } \sum_{i=1}^{\infty} \frac{\bar{e}_i^2}{e_i^{2c}} < \infty \right\}.$$

Assumption 6 *Let $c \in (1/2, 3/2]$. For any $\pi \in \Pi$ and any $Q \in \mathcal{F}$, we have $\mathcal{B}^\pi Q \in \mathcal{F} \subseteq \mathcal{H}_k^{2c}$. In addition, $\mathcal{F}$ is symmetric and closed under the limit operation and addition.*

The first statement of Assumption 6 is closely related to Bellman completeness, which has been widely imposed in the literature of batch RL (e.g., Antos et al., 2007). This assumption will typically hold when the transition and the reward functions are smooth enough. Since $\mathcal{F}$ is closed under the limit operator, Assumption 6 implies that $Q^\pi \in \mathcal{F}$ for $\pi \in \Pi$ by the contraction property of Bellman operator. In other words, there is no model misspecification error for estimating Q^π for every $\pi \in \Pi$. Moreover, together with symmetry and closed under addition, Assumption 6 also implies that for any $\pi \in \Pi$, any $Q \in \mathcal{F}$, $\mathcal{T}^\pi Q \in \mathcal{F} \subseteq \mathcal{H}_k^{2c}$. This further states that for any $\pi \in \Pi$, $Q \in \mathcal{F}$, there exists $g_{\pi,Q} \in L^2_{\bar{d}_T^\pi}(\mathcal{Z})$ such that $\mathcal{L}_k^c g_{\pi,Q} = \mathcal{T}^\pi Q$, where $\mathcal{L}_k^c$ is an operator associated with the kernel k and its definition can be found in Appendix 3. In the standard literature of the kernel ridge regression (e.g., Smale and Zhou, 2007; Caponnetto and De Vito, 2007), for fixed Q and π , the same type of the smoothness assumption is also imposed. $c \in (1/2, 3/2]$ is the range where the kernel ridge regression can adapt to the smoothness of the true function. Here we strengthen this smoothness assumption by considering for all $\pi \in \Pi$ and $Q \in \mathcal{F}$, which is used for controlling the bias uniformly over $\mathcal{F}$ and Π induced by the regularization in the kernel ridge regression (1) in the appendix. See Remark 2 below for more discussions. It is worth mentioning that one can choose $\mathcal{F} = \mathcal{H}_k^{2c}$ and modify Assumption 5 (a) accordingly.

4.2 Regret Bounds

To derive the regret bounds for $\hat{\pi}$ and $\hat{\pi}_{\text{dual}}$, we first show that for every $\pi \in \Pi$, Q^π is a feasible solution to the inner minimization problem of (21) with a high probability. Let $\Omega(\pi, \mathcal{F}, \mathcal{W}, \varepsilon_{NT}^{(1)}, \varepsilon_{NT}^{(2)}) \triangleq \Omega_1(\pi, \mathcal{F}, \mathcal{W}, \varepsilon_{NT}^{(1)}) \cap \Omega_2(\pi, \mathcal{F}, \varepsilon_{NT}^{(2)})$.

Theorem 1 *Let Assumptions 1-6, and Assumption 1 in Appendix hold. By letting*

$$\begin{aligned} \varepsilon_{NT}^{(1)} &\asymp \log(NT) \sqrt{(v(\Pi) + v(\mathcal{F})) / NT} \quad \text{and} \\ \varepsilon_{NT}^{(2)} &\asymp \left(\log(NT) \sqrt{(v(\Pi) + v(\mathcal{F}) + v(\mathcal{G})) / NT} \right)^{\frac{2c-1}{2c+1}}, \end{aligned}$$

with $c \in (1/2, 3/2]$ given in Assumption 6, we have, with probability at least $1 - 1/NT$, for every $\pi \in \Pi$,

$$Q^\pi \in \Omega(\pi, \mathcal{F}, \mathcal{W}, \varepsilon_{NT}^{(1)}, \varepsilon_{NT}^{(2)}). \quad (32)$$

Theorem 1 is the key for establishing the regret bounds of our proposed algorithms. By showing (32), we can guarantee that for every $\pi \in \Pi$, Q^π is bounded below by the optimal value of the inner minimization problem in (21) with a high probability. This verifies the use of pessimism in our algorithms, i.e., evaluating the value of each $\pi \in \Pi$ by a lower bound during the policy optimization procedure. For $c \in (1/2, 3/2]$, the smallest value that $\varepsilon_{NT}^{(2)}$ can be set is of the order $\sqrt{\log(NT)}(NT)^{-1/4}$. Based on Theorem 1, we have the following regret warranty for $\hat{\pi}$.

Theorem 2 (Regret warranty for $\hat{\pi}$) *Let Assumptions 1-6 and Assumption 1 in Appendix hold. Let $\varepsilon_{NT}^{(1)}$ and $\varepsilon_{NT}^{(2)}$ be given in Theorem 1. Then we have:*

(a) *Suppose $\omega^{\pi^*} \in \mathcal{W}$, then with probability at least $1 - \frac{1}{NT}$,*

$$\text{Regret}(\hat{\pi}) \lesssim \varepsilon_{NT}^{(1)} + \tilde{\lambda}_2^{\pi^*}(\mathcal{S} \times \mathcal{A}) \times \text{MMD}_k(\bar{d}_T^{\pi^*}, \lambda_2^{\pi^*}) \times \varepsilon_{NT}^{(2)}. \quad (33)$$

(b) Suppose $d_\gamma^{\pi^*} \ll \bar{d}_T^{\pi^b}$ and $\omega^{\pi^*} \in \mathcal{W}$, then with probability at least $1 - \frac{1}{NT}$,

$$\text{Regret}(\hat{\pi}) \lesssim \varepsilon_{NT}^{(1)}. \quad (34)$$

(c) Suppose $d_\gamma^{\pi^*} \perp \bar{d}_T^{\pi^b}$, then with probability at least $1 - \frac{1}{NT}$,

$$\text{Regret}(\hat{\pi}) \lesssim \varepsilon_{NT}^{(1)} + \text{MMD}_k(\bar{d}_T^{\pi^b}, \lambda_2^{\pi^*}) \times \varepsilon_{NT}^{(2)}. \quad (35)$$

Remark 1 Among all assumptions we have made, Bellman completeness (i.e., Assumption 6) and $\omega^{\pi^*} = \frac{\lambda_1^{\pi^*}}{\bar{d}_T^{\pi^b}} \in \mathcal{W}$ with $\|\omega^{\pi^*}\|_\infty < \infty$, are two main sufficient conditions imposed in Theorem 2. Assumption 6 is the smoothness modeling condition that is used to balance the MMD distance for the singularity part. assumption, while the second one is related to our batch data coverage induced by the behavior policy.

In the existing literature, the weakest data coverage assumption for establishing the regret guarantee is $d_\gamma^{\pi^*} \ll \bar{d}_T^{\pi^b}$ with $\|d_\gamma^{\pi^*} / \bar{d}_T^{\pi^b}\|_\infty < \infty$, i.e., absolute continuity with a uniformly bounded Radon–Nikodym derivative. See for example Xie et al. (2021) and Zhan et al. (2022). In particular, Xie et al. (2021) used the quantity $C(\pi) := \sup_{g \in \mathcal{G}} \|g - \mathcal{B}^\pi g\|_{d_\gamma^\pi} / \|g - \mathcal{B}^\pi g\|_{\bar{d}_T^{\pi^b}}$ to quantify the distributional shift. But in the proof of their Theorem 3.1, they actually required the absolute continuity. In contrast, our proposed method remains valid without requiring the absolute continuity. For example, in Case (a) of Theorem 2, to achieve the regret guarantee of our proposed method, we only require the ratio of the absolutely continuous part of $d_\gamma^{\pi^*}$ with respect to $\bar{d}_T^{\pi^b}$ over $\bar{d}_T^{\pi^b}$, i.e., $\frac{\lambda_1^{\pi^*}}{\bar{d}_T^{\pi^b}}$, is uniformly bounded above, and the MMD between the singular part of $d_\gamma^{\pi^*}$ with respect to $\bar{d}_T^{\pi^b}$ and $\bar{d}_T^{\pi^b}$, i.e., $\text{MMD}_k(\bar{d}_T^{\pi^b}, \lambda_2^{\pi^*})$, is finite, which is ensured by Assumption 1. Therefore our method requires much weaker assumptions (essentially no assumptions) on the batch data coverage induced by the behavior policy, and is thus more applicable than existing algorithms from the perspective of the data coverage. However, our method requires a stronger condition on modeling Q^π , which could be regarded as a trade-off. We require Bellman completeness and symmetric, additive

functional class for modeling Q^π (i.e., Assumption 6), while some existing pessimistic algorithms (e.g., Jiang and Huang, 2020) only require the realizability condition on modeling Q^π (i.e., there is no misspecification error for Q^π).

It is worth mentioning that, when $d_\gamma^{\pi^*} \ll \bar{d}_T^{\pi^b}$ as discussed in the Case (b) of Theorem 2, ω^{π^*} becomes $d_\gamma^{\pi^*}/\bar{d}_T^{\pi^b}$, with $\tilde{\lambda}_2^{\pi^*}(\mathcal{S} \times \mathcal{A}) = 0$. In this case, we recover the existing results such as Jiang and Huang (2020) on the regret bound by running our algorithm. However, compared with their results, we require a stronger condition, i.e., Assumption 6 for modeling Q^π instead of realizability condition on Q^π only. The main reason for requiring a stronger condition here is due to the unknown information that $d_\gamma^{\pi^*} \ll \bar{d}_T^{\pi^b}$. When $d_\gamma^{\pi^*} \perp \bar{d}_T^{\pi^b}$ as discussed in Case (c) of Theorem 2, it can be seen that $\tilde{\lambda}_2^{\pi^*}(\mathcal{S} \times \mathcal{A}) = 1$. We showcase that the regret of our estimated policy can still possibly converge to 0 in a satisfactory rate without any data coverage assumption. To achieve this, we rely on the Bellman completeness and the smoothness condition for enabling the extrapolation property. To the best of our knowledge, this is the first regret guarantee in batch RL without assuming the absolute continuity. See Table 1 in Appendix for a summary of our main assumptions for obtaining the regret of the proposed algorithm compared with existing ones. Under the present smoothness assumption $c > 1/2$, the term $\varepsilon_{NT}^{(2)}$ vanishes and the regret bound can converge to zero.

Remark 2 Our proposed algorithm is motivated by Lemma 2, which provides a careful decomposition on the estimation error of OPE without relying on the absolute continuity. The key reason for the possible success of our algorithm in handling insufficient data coverage (i.e., the singular part) relies on the smoothness condition we imposed on modeling Q^π , i.e., Assumption 6, which provides a strong extrapolation property for our estimated $\hat{T}^\pi Q$ for $Q \in \mathcal{F}$ and $\pi \in \Pi$. Specifically, for any state-action pair $(s, a) \in F$, where F is a measurable set such that $\bar{d}_T^{\pi^b}(F) = 0$ but $\bar{d}_T^{\pi^b}(F^c) = 1$, by leveraging the convolution operator $\mathcal{L}_k$ defined in Appendix 3, the corresponding $\mathcal{T}^\pi Q(s, a)$ can be extrapolated by the kernel

smoothing of $\mathcal{T}^\pi Q(\tilde{s}, \tilde{a})$ for all $(\tilde{s}, \tilde{a}) \in F^c$. Therefore, $\hat{\mathcal{T}}^\pi Q$ can approximate $\mathcal{T}^\pi Q$ well in terms of the RKHS norm. Such an extrapolation property is inherited by Q -function due to the Bellman equation that $\mathcal{T}^\pi Q^\pi = 0$ (i.e., $\mathcal{B}^\pi Q^\pi = Q^\pi$).

Next, we develop a finite-sample regret bound for $\hat{\pi}_{\text{dual}}$.

Theorem 3 *Under Assumptions 1-6 and Assumption 1 in Appendix, all statements in Theorem 2 hold for $\hat{\pi}_{\text{dual}}$ provided $\mathcal{F}$ is a convex set.*

Remark 3 *The implications behind Theorem 3 are the same as those of Theorem 2. The benefit of considering $\hat{\pi}_{\text{dual}}$ mainly comes from the computational efficiency. However, assuming $\mathcal{F}$ convex could be restrictive. Therefore it will be interesting to study the behavior of duality gap when $\mathcal{F}$ is modeled by a non-convex but “asymptotically” convex set. This will allow the use of some deep neural network architectures such as Zhang et al. (2019) in practice. We leave it for future work.*

5 Simulation Studies

In this section, we use Monte Carlo (MC) experiments to systematically study the proposed method in terms of the convergence of our algorithm, the effect of pessimism in finding the optimal policy, and the robustness to the covariate shift. For simplicity, we only consider the contextual bandit problem.

For all numerical studies, we compare our method with two popular continuous-action policy optimization methods. The regression-based approach first estimates the reward function $r(s, a)$ as $\hat{r}(s, a)$ by running a regression method, and then estimates the value of any given policy π with the plug-in estimator $\mathbb{E}_{S_0 \sim \nu} [\hat{r}(S_0, \pi(S_0))]$, where the expectation is approximated via MC sampling of ν . Recall that ν is known in advance. Given such a policy value estimator, we implement the off-the-shelf optimization algorithm to estimate

the optimal policy within a given policy class. The kernel-based approach (Chen et al., 2016; Kallus and Zhou, 2018) is similar to the regression approach but first estimates the policy value using the inverse probability weighting with kernel smoothing. Specifically, we estimate the value of any given policy π by $(Nh)^{-1} \sum_{i=1}^N \frac{K((\pi(S_{i,0}) - A_{i,0})/h)}{\pi^b(A_{i,0} | S_{i,0})} \times R_{i,0}$, where $K(\cdot)$ is a kernel function, h is the bandwidth parameter, and π^b is referred as to the generalized propensity score in this setting.

Next, we describe the data generating process in our simulation study. We consider the mean reward function $r(s, a) = (a - \mathbf{B}s)^\top \mathbf{C}(a - \mathbf{B}s)$, where the state-value s has dimension $d_s = 5$, the action-value a has dimension $d_a = 4$, the parameter matrix $\mathbf{B} \in \mathcal{R}^{d_a \times d_s}$, and $\mathbf{C}$ is a $d_a \times d_a$ negative definite matrix. We randomly sample every entry of $\mathbf{B}$ from $\text{Uniform}(0, 1)$. We construct $\mathbf{C} = -\mathbf{C}_0^\top \mathbf{C}_0$, where every entry of $\mathbf{C}_0$ is sampled from the standard normal. Our reward R_0 is generated following $R_0 = r(S_0, A_0) + \epsilon_0$, where the independent noise $\epsilon_0 \sim \mathcal{N}(0, 1)$. Therefore, the optimal policy $\pi^*(s) = s^\top \mathbf{B}$ for every $s \in \mathcal{S}$, regardless of the state distribution.

The training dataset $\{S_{i,0}, A_{i,0}, R_{i,0}\}_{1 \leq i \leq N}$ is generated as follows. The state $S_{i,0}$ is uniformly sampled from $[0, 2]^{d_s}$, which is the same as the reference distribution. Except for the case where we study the effect of covariate shift, values of learned policies in this section are evaluated over the same distribution as the training distribution. The action is sampled following a behavior policy π^b such that $A_{i,0} = \mathbf{B}S_{i,0} + \epsilon'$, where $\epsilon' \sim \mathcal{N}(0, \sigma_b^2 \mathbf{I}_{d_a})$. Therefore, a larger σ_b indicates that the behavior policy is more different from the optimal one and also implies that the action space is explored more in the training data.

For all three methods, we search the optimal policy within the class of linear deterministic policies $\Pi = \{\pi \mid \text{for every } s \in \mathcal{S}, \pi(s) = \tilde{\mathbf{B}}s \text{ for some } \tilde{\mathbf{B}} \in [-1, 1]^{d_s \times d_a}\}$. For the kernel-based method, we assume the true value of the generalized propensity score π^b is *known*, instead of plugging-in the estimated value. Although the latter is known to be

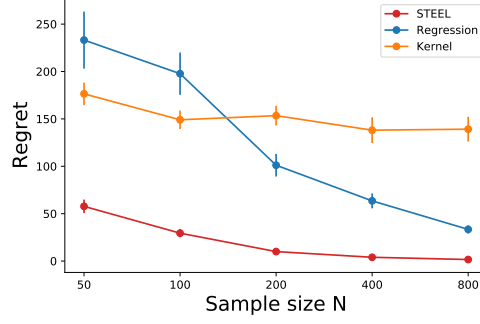


Figure 1: Convergence of different methods. The error bars indicate the standard errors of the averages. The x-axis is plotted on the logarithmic scale.

asymptotically more efficient in the causal inference literature, we observe that its empirical performance is close or sometimes worse than the oracle version in the studied setting, given the challenge to estimating a multi-dimensional conditional density function (See Appendix 7). We tune the bandwidth for the kernel-based method and use the best one for it. To implement the optimization problems in two baseline methods, we adopt the widely used L-BFGS-B optimization algorithm following [Liu and Nocedal \(1989\)](#). Regarding estimating the reward function $r(s, a)$, we use a neural network of three hidden layers with 32 units for each layer and ReLU as the activation function. For **STEEL**, we use the Laplacian kernel $k(x, y) = \exp(-\|x - y\|_1/\gamma)$ in both the kernel ridge regression and the modeling of ω , where the bandwidth γ is picked based on the popular median heuristic, i.e., set as the median of $\{\|(S_{i,0}, A_{i,0}) - (S_{j,0}, A_{j,0})\|_1\}_{1 \leq i, j \leq N}$. For the other tuning parameters, we set $\zeta_{NT} = 0.001$, $(\varepsilon_{NT}^{(1)})^2 = 300$, and $(\varepsilon_{NT}^{(2)})^2 = 600$.

For each setting below, we run 100 repetitions and report the average regret of each method as well as its standard error. Except for when we study the convergence of three methods in terms of the sample size, we fix $N = 200$ data points for each repetition.

Convergence. We first study the convergence of all three methods, by increasing the training sample size N . The results when $\sigma_b = 0.5$ are presented in Figure 1. Overall, we

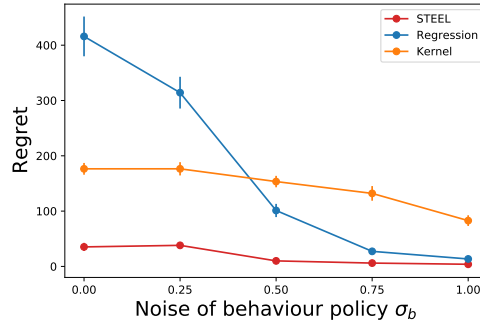


Figure 2: Effect of pessimism. The error bars indicate the standard errors of the averages.

observe that our method yields a desired convergence with the regret decaying to zero very quickly. The regression-based method also improves as the sample size increases, despite that the finite-sample performance is worse compare with ours. The kernel-based method suffers from the curse of dimensionality of the action space and the regret does not decay very significantly. Results under a few other settings are reported in the appendix, with similar findings.

Effect of pessimism. Next recall that σ_b controls the degree of exploration in the training data. It is known that a well-explored action space in the training dataset is beneficial for policy learning, in the sense that we can accurately estimate the values of most candidate policies. In contrast, when σ_b is small (i.e., the behavior policy is close to the optimal one), although the value estimation of the optimal policy may be accurate, that of a sub-optimal policy suffers from a large uncertainty and by chance the resulting estimator could be very large, which leads to returning a sub-optimal policy. Our method is designed for addressing such a large uncertainty, by utilizing the pessimistic mechanism.

In Figure 2, we empirically study the effect of σ_b on different methods. The results are based on $N = 200$ data points and different values of σ_b . It can be seen clearly that, our method is fairly robust, when the training data is either well explored or not. In contrast, the performance of the two baseline methods deteriorates significantly when σ_b is small.

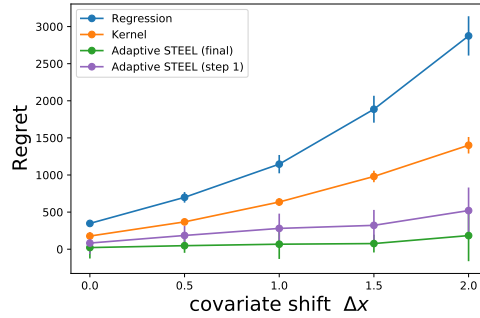


Figure 3: Robustness to covariate shift. The error bars indicate the standard errors.

Performance of adaptive STEEL. In this section, we numerically study the performance of the adaptive version of STEEL, i.e., Algorithm 2. Notice that the singularity can be caused by the covariate shift and the distance between the data and the target distributions will also increase with covariate shift, we design such a setting for our study. Specifically, the policy will be tested when the covariates are sampled uniformly from $[\Delta x, 2 + \Delta x]^{d_s}$. A larger value of Δx hence indicates a larger degree of covariate shift. We set a very large initial $\varepsilon_{NT}^{(2)} = 5000$, and report the performance of both the first-stage policy and the final policy from Algorithm 2.

The results when $\sigma_b = 0.25$ and $N = 200$ are presented in Figure 3. It can be seen clearly that adaptive STEEL, without further tuning, consistently shows robust and better performance than other algorithms. The first-stage policy from Algorithm 2, although still has decent performance, shows higher regrets that grow with the degree of covariate shift due to its lack of adaptiveness from the appropriately constructed uncertainty set.

6 Real Data Application

In this section, we apply our method to personalized pricing with a real dataset. The dataset is from an online auto loan lending company in the United States¹. It was first studied by Phillips et al. (2015) and we follow similar setups as in subsequent studies (Ban

¹The dataset is available at <https://business.columbia.edu/cprm> upon request.

and Keskin, 2021).

The dataset contains all auto loan records (208085 in total) in a major online auto loan lending company from July 2002 to November 2004. For each record, it contains covariates (e.g., FICO credit score, the term and amount of loan requested, etc.), the monthly payment offered by the company, and the decision of the applicant (accept the offer or not). We refer interested readers to Phillips et al. (2015) for more details of the dataset.

The monthly payment (together with the loan term) can be regarded as the offered price (*action*) and the binary decision of the applicant reflects the corresponding demand. Specifically, we follow Ban and Keskin (2021) and Bastani et al. (2022) to define the price as $\text{price} = \text{Monthly payment} \times \sum_{t=1}^{\text{Term}} (1 + \text{Prime rate})^{-t} - \text{amount of loan requested}$, which considers the net present value of future payments. The prime rate is the interest rate that the company needs to pay. We filter the dataset to exclude the outlier records (defined as having a price higher than \$10,000), and 99.49% data points remain in the final dataset.

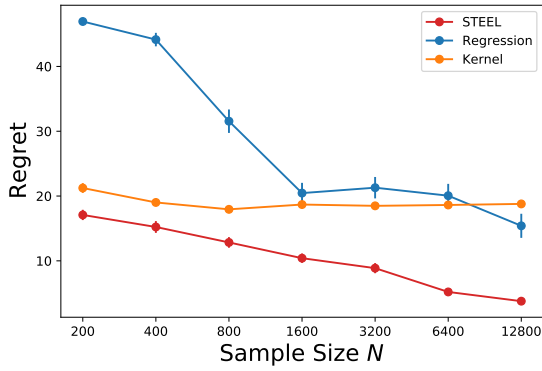
In our notations, the price is hence the action A_0 . Let the binary decision of the applicant be D_0 . The reward is then naturally computed as $R_0 = D_0 \times A_0$. We use the same set of features which are identified as significant in Ban and Keskin (2021) to construct our feature vector S_0 : it contains the FICO credit score, the loan amount approved, the prime rate, the competitor’s rate, and the loan term. When running offline experiments for problems with *discrete* action spaces, to compared different learned policies, the standard procedure is to find those data points where the recorded actions are consistent with the recommendations from a policy π that we would like to evaluate. However, with a *continuous* action space, such a procedure is typically impossible, as it is difficult for two continuous actions to have exactly the same value. Therefore, it is well acknowledged that, offline experiments without any model assumption is infeasible. We closely follow Bastani et al. (2022) to design a semi-real experiment, where we fit a linear demand function from

the real dataset and use it to evaluate the policies learned by different methods. The linear regression uses the concatenation of S_0 (which describes the baseline effect) and S_0A_0 (which describes the interaction effect) as the covariate vector. To evaluate the statistical performance of the three methods considered in Section 5, we repeat for 100 random seeds, and for each random seed we sample N data points for policy optimization.

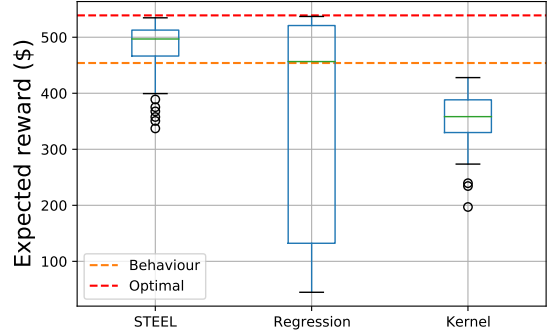
The implementation and tuning details of the three methods are almost the same with Section 5. The main modification is that, for the reward function class used in our method and the regression-based method, we use the neural network to model the demand function (instead of the expected reward itself), which multiplies with the action (price) gives the expected reward. Such a modification utilizes the problem-specific structure. All covariates are normalized and we always add an intercept term. For tuning parameters, we set $\zeta_{NT} = 0.001$, $(\varepsilon_{NT}^{(1)})^2 = 13$, and $(\varepsilon_{NT}^{(2)})^2 = 16$.

In Figure 4a, we increase N from 200 to 12800 and report the average regrets. We find that our method consistently outperforms the other two methods and its regret decays to zero. In Figure 4b, we zoom into the case where $N = 8000$, and report the values of the learned policies across the 100 random seeds. We observe that, although the regression-based method can outperform the behaviour policy in about half of the random replications, it can perform very poor in the remaining replicates. In contrast, our method has an impressive performance and the robustness of our methods is consistent with our methodology design. The kernel-based method does not perform well in most settings. A deep dive shows that this is because this method over-estimates the values of some sub-optimal policies that assign many actions near the boundaries where we have few data points. The over-estimation is due to its poor extrapolation ability and the lack of the pessimism mechanism (i.e., it is not aware of the uncertainty).

Finally, we randomly pick a policy learned by our method (with the first random seed



(a) Regret trend with sample size. The error bars indicate the standard errors of the averages.



(b) Value distribution over 100 seeds when $N = 8000$.

Figure 4: Performance in personalized pricing with real data.

Table 1: Coefficients of the learned linear personalized pricing policy.

FICO score	Loan amount approved	Prime rate	Competitor's rate	Term
-0.135	0.041	0.180	0.027	0.514

we used) and report its coefficient to provide some insights. Recall that we use a linear policy, mapping from the five features to the recommended price. The coefficients are overall aligned with the intuition: the learned personalized pricing policy offers a higher price when the FICO score is lower, the loan amount approved is higher, the prime rate (the interest rate that the company itself faces) is higher, the competitor's rate is higher (implying a consensus on the high risk of this loan), and the term is longer.

7 Conclusion

In this paper, we study batch RL in the presence of singularities and propose a new policy learning algorithm to tackle this issue. Our proposed method finds an optimal policy without requiring absolute continuity on the distribution of target policies with respect to

the data distribution. Both theoretical guarantees and numerical evidence are presented to illustrate the superior performance of our proposed method, compared with existing works. In the current proposal, we leverage the MMD, together with distributionally robust optimization, to enable the extrapolation for addressing the singularity. Our idea can also be extended to other distances such as the Wasserstein distance or total variation distance. While these distances have some appealing properties, the related estimation becomes difficulty, showing a trade-off. Finally, it is also interesting to explore model-based solutions for addressing the singularity issue in batch RL.

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